Day-03_Slides

January 20, 2026

CMSE 801 - Day 3 ## Calculating savings growth using lists and loops ### Jan 20, 2026

0.1 General Course Announcements

- The due dates for Homework assignments and dates for the quizzes will be finalized by next week and updated on the course schedule page.
- Office hours and help room hours have started this week! **You can find the schedules times and location on the course homepage.**
- Remember: at the end of class **you will be uploading your in-class notebook to D2L**.

0.2 Questions/comments from Day 3 PCA

- Some confusing terminology and jargon coming up here and there. e.g. things "initialize" and "iterate"
- Some challenges with understanding what exactly a question is asking you to do.
- When/why would I use a nested loop?
- What are the key differences between `for` loops and `while` loops?
  - **We'll spend a bit of time talking/thinking about this**
- Some struggles with Question 5 and Question 6
  - **We'll take a look at some possible solutions and some potential pitfalls**

0.3 The `for` loop vs. the `while` loop

What's the key difference?

0.4 The `for` loop vs. the `while` loop

0.5 Let's look at some code examples

[11]:
```
# Loop by integer values or "by index"
for i in range(0,3):
    print(i)
```

```
0
1
2
```

[13]:
```
# Loop by value in a list
mylist = ['red', 'yellow', 'blue']
for value in mylist:
```

```
    print(value)
```

```
red
yellow
blue
```

[9]:
```
# Loop until the the "count" value is greater than or equal to 3
count = 0
while count < 3:
    print(count)
    count += 1
```

```
0
1
2
```

0.6 Reviewing Question 5

Below is a possible solution to Question 5, will it work? If not, what's missing?

[14]:
```
x = [2,4,6,8,10,12,14,16,18]
y = [10,8.25,7.5,7,6.5,7,7.5,8.25,10]

# Create a loop to add the x and y values together and store them in a list␣
 ↪called "addition"
for i in range(len(x)):
    # index the x and y lists, add their values, and append them to the list
    addition.append(x[i] + y[i])

# Check to see if it worked
print(addition)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[14], line 7
      4 # Create a loop to add the x and y values together and store them in a␣
 ↪list called "addition"
      5 for i in range(len(x)):
      6     # index the x and y lists, add their values, and append them to the␣
 ↪list
----> 7     addition.append(x[i] + y[i])
      9 # Check to see if it worked
     10 print(addition)

NameError: name 'addition' is not defined
```

This should work!

[15]:
```python
x = [2,4,6,8,10,12,14,16,18]
y = [10,8.25,7.5,7,6.5,7,7.5,8.25,10]

# Initialize my empty list
addition = []

# Create a loop to add the x and y values together and store them in a list␣
↪called "addition"
for i in range(len(x)):
    # index the x and y lists, add their values, and append them to the list
    addition.append(x[i] + y[i])

# Check to see if it worked
print(addition)
```

```
[12, 12.25, 13.5, 15, 16.5, 19, 21.5, 24.25, 28]
```

0.7 Reviewing Question 6 (the nested loop!)

What if for each individual `x` value, we want to calculate $(x + y)^2$ for *all* of the y values and print the list for each `x` value?

[16]:
```python
x = [2,4,6,8,10,12,14,16,18]
y = [10,8.25,7.5,7,6.5,7,7.5,8.25,10]

# Time to nest some loops!
for i in range(len(x)):
    xy_squared = []
    # index the x and y lists, add their values, and append them to the list
    for ii in range(len(y)):
        xy_squared.append((x[i] + y[ii])**2)

    # Check to see if it worked
    print(xy_squared)
```

```
[144, 105.0625, 90.25, 81, 72.25, 81, 90.25, 105.0625, 144]
[196, 150.0625, 132.25, 121, 110.25, 121, 132.25, 150.0625, 196]
[256, 203.0625, 182.25, 169, 156.25, 169, 182.25, 203.0625, 256]
[324, 264.0625, 240.25, 225, 210.25, 225, 240.25, 264.0625, 324]
[400, 333.0625, 306.25, 289, 272.25, 289, 306.25, 333.0625, 400]
[484, 410.0625, 380.25, 361, 342.25, 361, 380.25, 410.0625, 484]
[576, 495.0625, 462.25, 441, 420.25, 441, 462.25, 495.0625, 576]
[676, 588.0625, 552.25, 529, 506.25, 529, 552.25, 588.0625, 676]
[784, 689.0625, 650.25, 625, 600.25, 625, 650.25, 689.0625, 784]
```

0.8 Goals for today

- Get some practice and comfort with writing loops and using lists in Python.
- Use these tools to track accumulated savings over time using a retirement investment model.

0.9 Looking forward:

0.9.1 Exploring conditional statements and a brief intro to functions in the Day 4 PCA

- "`If`" statments in Python (and their more complex "`elif`" and "`else`" statements)
  - Used to control code flow
- Functions in Python
  - Modular
  - Reusable
  - Useful!

Note: this might also feel like a slightly large PCA, but you have a few days to work though it, I encourage you to tackle it sooner than later.